

1. To study Configuration of IP and MAC Address

Aim:

To study the configuration of IPv4 Address and MAC Address and verify network connectivity.

Q1. Steps:

1. Press **Windows + R**
2. Type **cmd** and press Enter
3. Type: **ipconfig /all**

Note down:

- IPv4 Address: _____
 - Subnet Mask: _____
 - Default Gateway: _____
 - Physical (MAC) Address: _____
-

Q2. Write down the steps to configure new IP address on your PC.

1. Open **Control Panel**
 2. Click **Network and Internet**
 3. Open **Network and Sharing Center**
 4. Click **Change Adapter Settings**
 5. Right-click **Ethernet/Wi-Fi**
 6. Select **Properties**
 7. Select **Internet Protocol Version 4 (TCP/IPv4)**
 8. Click **Properties**
 9. Select **Use the following IP address**
 10. Enter: IP Address, Subnet Mask, Default Gateway
 11. Click **OK**
-

Q3. Configure your PC with new IPv4 Address

Example:

- IPv4 Address: **192.168.1.100**
 - Subnet Mask: **255.255.255.0**
 - Default Gateway: **192.168.1.1**
-

Q4. Verify the configuration and check connectivity. Note down observations.

Command:

ping 192.168.1.1

Observations:

- IPv4 address and MAC address were displayed successfully.
 - New IP configuration was applied successfully.
 - The system responded to the ping command.
 - Network connectivity was established successfully.
 - No packet loss was observed.
-

Q5. What is the difference between Logical Address and Physical Address?

Logical Address	Physical Address
IP Address	MAC Address
Can be changed	Usually fixed
Used for routing	Used for device identification
Assigned by network administrator. Assigned by manufacturer	

Q6. State the use of ping command.

Answer:

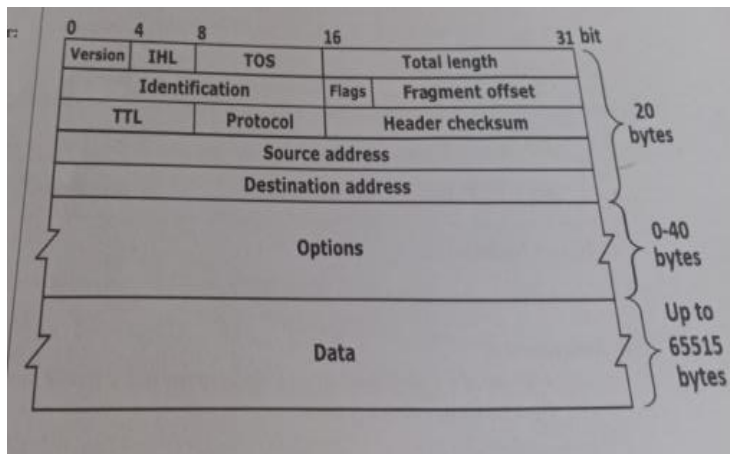
The **ping command** is used to test communication between two network devices and verify whether the destination host is reachable.

3. To convert the given IP addresses from dotted-decimal notation to binary notation

Aim:

To study IPv4 addressing and convert IPv4 addresses from dotted-decimal notation into binary notation.

Q1. Draw the IPv4 header format.



Version – IP version number.
IHL – Header length. **TOS** – Type of service. **Total Length** – Total packet size. **Identification** – Identifies packet fragments.
Flags – Controls fragmentation. **Fragment Offset** – Position of fragment.
TTL – Packet lifetime limit.
Protocol – Type of protocol used.

Header Checksum – Checks header errors.

Source Address – Sender IP address. **Destination Address** – Receiver IP address.

Options – Extra control information. **Data** – Actual message/data.

Q2. Turn on the PC, find and note down the IPv4 Address, Subnet Mask, Default Gateway and Physical (MAC) Address. Show it to examiner.

Steps:

1. Press **Windows + R**
2. Type **cmd**
3. Enter:

ipconfig /all

Note:

- IPv4 Address: _____
- Subnet Mask: _____

- Default Gateway: _____

- Physical (MAC) Address: _____

Observations:

- IPv4 addresses were identified successfully.
- Decimal IPv4 addresses were converted into binary notation.
- Invalid IP addresses were detected correctly

Q3. Convert the IPv4 address of your PC into binary notation.

Example:

IPv4 Address: **192.168.1.100**

Binary:

- 192 → **11000000**
- 168 → **10101000**

- 1 → **00000001**

- 100 → **01100100**

Binary IPv4:

11000000.10101000.00000001.01100100

Q4. Convert the given IPv4 addresses into binary notation and identify wrong IPv4 address if any.

a) **148.162.1.12**

- 148 → 10010100
- 162 → 10100010
- 1 → 00000001
- 12 → 00001100

Binary:

10010100.10100010.00000001.00001100

c) **127.256.0.1**

Wrong IPv4 Address ✕

(256 exceeds maximum value of 255)

d) **12.11.22.2**

- 12 → 00001100
- 11 → 00001011
- 22 → 00010110
- 2 → 00000010

Binary:

00001100.00001011.00010110.00000010

b) **258.2.1.22**

Wrong IPv4 Address ✕

(258 exceeds maximum value of 255)

e) IPv4 Address: _____, Binary: _____

Q5. What is the range of value for each octet in an IPv4 address?

Each octet value ranges from **0 to 255**.

Q6. Write the total address space of IPv4 address.

Answer:

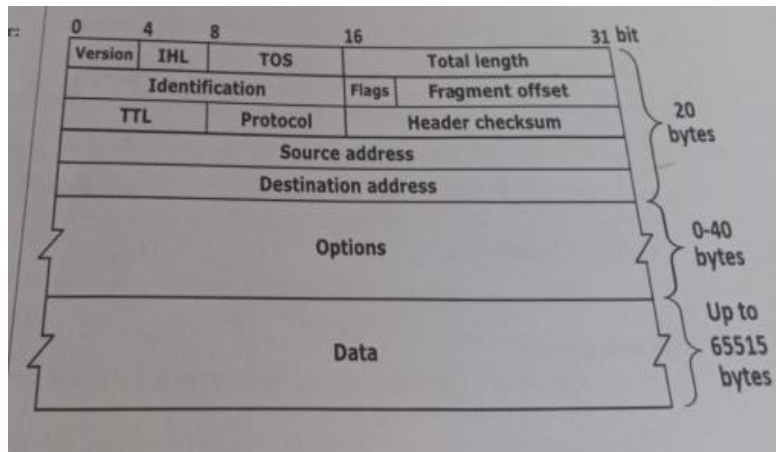
Total IPv4 address space = 2^{32} = **4,294,967,296** addresses

4. To convert the given IP addresses from binary to dotted-decimal notation

Aim:

To study IPv4 addressing and convert binary IPv4 addresses into dotted-decimal notation.

Q1. Draw the IPv4 header format.



Version – IP version number.

IHL – Header length. **TOS** –

Type of service. **Total**

Length – Total packet size.

Identification – Identifies packet fragments. **Flags** –

Controls fragmentation.

Fragment Offset – Position of fragment. **TTL** – Packet lifetime limit.

Protocol – Type of protocol

used. **Header Checksum** – Checks header errors.

Source Address – Sender IP address. **Destination Address** – Receiver IP address.

Options – Extra control information. **Data** – Actual message/data.

Q2. Turn on the PC, find and note down the IPv4 Address, Subnet Mask, Default Gateway and Physical (MAC) Address. Show it to examiner.

Steps: 1. Press **Windows + R**

2. Type **cmd**

3. Enter: **ipconfig /all**

Note:

- IPv4 Address:

- Subnet Mask:

- Default Gateway:

- Physical (MAC) Address:

Observations:

- Binary IPv4 addresses were converted successfully into dotted-decimal notation.
- Invalid IPv4 addresses were identified correctly.
- IPv4 consists of 32 bits divided into 4 octets.

Q3. Change the given IPv4 address into dotted-decimal notation.

a)
11000000.10101000.00000001.00001010

- 11000000 = 192
- 10101000 = 168
- 00000001 = 1
- 00001010 = 10

Dotted Decimal:
192.168.1.10

b)
01111111.00000000.00000000.00000000

- 01111111 = 127
- 00000000 = 0
- 00000000 = 0
- 00000000 = 0

Dotted Decimal:
127.0.0.0

c)

Binary Address: _____

Dotted Decimal: _____

Q4. Find and comment on the error if any in the following IP addresses.

a) **111.56.045.78**

✗ Invalid — leading zero (**045**) is not generally used in IPv4 notation.

b) **11100010.23.14.55**

✗ Invalid — first octet is in binary form while others are in decimal.

c) **221.34.7.45.8**

✗ Invalid — IPv4 address contains **5 octets** instead of 4.

d) **45.75.301.14**

✗ Invalid — **301 exceeds maximum value of 255.**

Q5. How many bits are there in an IPv4 address?

IPv4 address contains **32 bits**.

Q6. Write the total address space of IPv4 address.

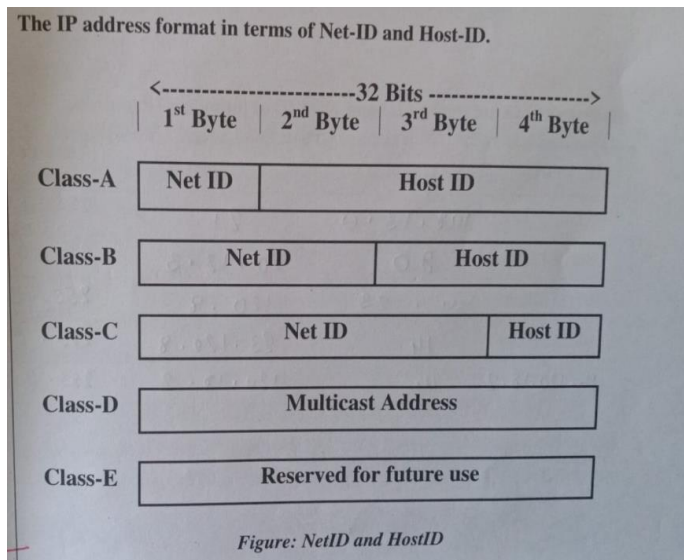
Total IPv4 address space = $2^{32} = 4,294,967,296$ addresses

5. To find Net-ID and Host-ID of the given IP addresses

Aim:

To study IPv4 addressing and identify the Net-ID and Host-ID of given IP addresses.

Q1. Draw the IPv4 address format in terms of Net-ID and Host-ID.



Class A – Net ID = 1 byte, Host ID = 3 bytes.

Class B – Net ID = 2 bytes, Host ID = 2 bytes.

Class C – Net ID = 3 bytes, Host ID = 1 byte.

Class D – Used for multicast addresses.

Class E – Reserved for future use.

Net ID – Identifies the network.

Host ID – Identifies the device in the

network.

Q2. Turn on the PC, find and note down the IPv4 Address, Subnet Mask, Default Gateway and Physical (MAC) Address. Show it to examiner.

Steps:

1. Press **Windows + R**
2. Type **cmd**
3. Enter:

ipconfig /all

Note:

- IPv4 Address: _____
- Subnet Mask: _____
- Default Gateway: _____
- Physical (MAC) Address: _____

Q3. Identify the Net-ID and Host-ID of the same.

Example:

IPv4 Address: **192.168.1.100**

Subnet Mask: **255.255.255.0**

- Net-ID: **192.168.1.0**
 - Host-ID: **100**
-

Q4. For IP address 192.168.2.130/25, find the Net-ID and Host-ID.

IP Address: **192.168.2.130/25**

/25 means subnet mask = **255.255.255.128**

- Net-ID: **192.168.2.128**
 - Host-ID: **2**
-

Q5. Define Net-ID and Host-ID.

Net-ID:

The part of an IP address used to identify a network.

Host-ID:

The part of an IP address used to identify a device within the network.

Q6. How many usable hosts are there in a /28 network?

For /28:

- Total host bits = **4**
- Total addresses = $2^4 = 16$
- Usable hosts = **16 - 2**

Answer: 14 usable hosts

Observations:

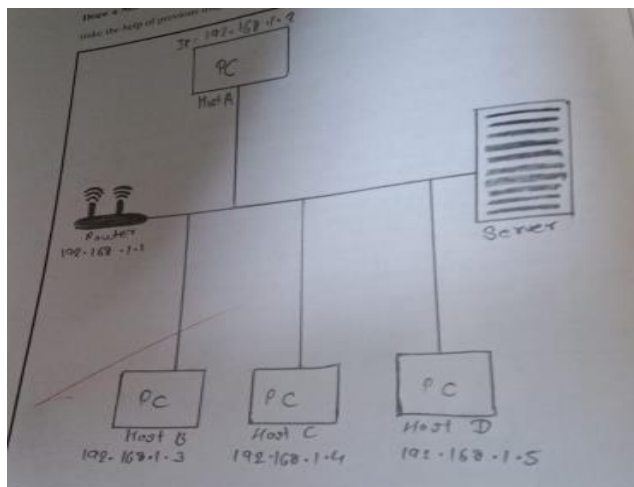
- IPv4 configuration details were obtained successfully.
- Net-ID and Host-ID were identified correctly.
- Subnet mask determines network and host portions.
- A /28 network provides 14 usable host addresses.

6. To find class of the given IP addresses

Aim:

To study IPv4 address classes and identify the class of given IP addresses.

Q1. Draw the network connecting at least 4 PCs with any class of IP address Mention IP



Q2. Turn on the PC, find and note down the IPv4 Address, Subnet Mask, Default Gateway and Physical (MAC) Address. Show it to examiner.

Steps:

1. Press **Windows + R**
2. Type **cmd**
3. Enter: **ipconfig /all**

Note:

- IPv4 Address:

- Subnet Mask:

- Default Gateway:

- Physical (MAC) Address:

Observations:

- IPv4 details of the PC were obtained successfully.
 - Different IP classes were identified using the first octet.
 - Default subnet masks for Class A, B and C were studied.
-

Q3. Find the class of your PC's IP address. Example:

IPv4 Address = **192.168.1.100**, Since first octet = **192** ,Class = **C**

Q4. What are the start and end addresses for different classes of IP address? Find class of following IP addresses.

Class	Start Address	End Address
A	1.0.0.0	126.255.255.255
B	128.0.0.0	191.255.255.255
C	192.0.0.0	223.255.255.255
D	224.0.0.0	239.255.255.255
E	240.0.0.0	255.255.255.255

Find class:

- a) **10.5.25.7** → **Class A**
 - b) **192.168.1.100** → **Class C**
 - c) **203.5.113.25** → **Class C**
-

Q5. Which bits determine the class of an IP address?

Answer:

The **first bits of the first octet** determine the class of an IP address.

- Class A → **0**
 - Class B → **10**
 - Class C → **110**
 - Class D → **1110**
 - Class E → **111**
-

Q6. What is default subnet mask of Class A, B and C?

Class Default Subnet Mask

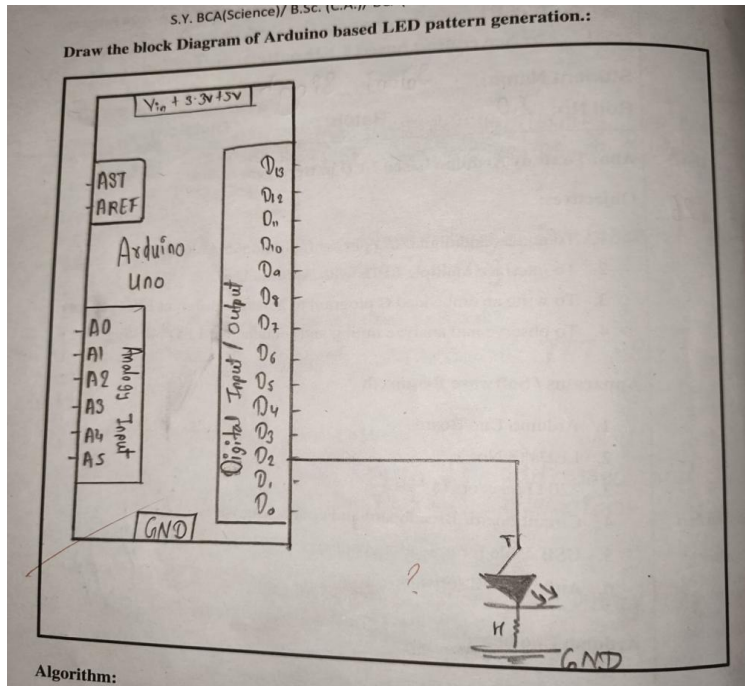
- A 255.0.0.0
- B 255.255.0.0
- C 255.255.255.0

7. To study Arduino based LED pattern generation

Aim:

To study LED pattern generation using Arduino and execute different LED blinking patterns.

Q1. Draw the circuit / block diagram of LED pattern generation using Arduino.



Circuit Description:

Connect multiple LEDs (e.g., 4 LEDs) to Arduino digital pins (e.g., pins 2, 3, 4, 5) Each LED connected in series with a 220Ω resistor to limit current
Common GND to Arduino GND

Observations:

- Arduino board interfaced successfully with LEDs.
- Program compiled and uploaded without errors.
- LEDs generated the desired blinking pattern.

Different patterns were created by modifying the program

Q2. Write algorithm / flowchart for same.

Algorithm:

1. Start
2. Initialize LED pins as OUTPUT
3. Turn ON LED1
4. Wait for some time
5. Turn OFF LED1 and turn ON LED2
6. Repeat for all LEDs
7. Continue pattern in loop
8. Stop

Q3. Write program and get it checked from the examiner.

```
int led1=2;
```

```
int led2=3;
```

```
int led3=4;
```

```
int led4=5;
```

```
void setup()
```

<pre> { pinMode(led1,OUTPUT); pinMode(led2,OUTPUT); pinMode(led3,OUTPUT); pinMode(led4,OUTPUT); } void loop() { digitalWrite(led1,HIGH); delay(500); digitalWrite(led1,LOW); </pre>	<pre> digitalWrite(led2,HIGH); delay(500); digitalWrite(led2,LOW); digitalWrite(led3,HIGH); delay(500); digitalWrite(led3,LOW); digitalWrite(led4,HIGH); delay(500); digitalWrite(led4,LOW); } </pre>
--	---

Q4. Execute the program using Arduino IDE. Show the output to the examiner.

Output:

LEDs glow one after another in sequence repeatedly.

Q5. Modify the program for different LED pattern generation.(Alternate LED pattern)

<pre> void loop() { digitalWrite(led1,HIGH); digitalWrite(led3,HIGH); digitalWrite(led2,LOW); digitalWrite(led4,LOW); </pre>	<pre> delay(500); digitalWrite(led1,LOW); digitalWrite(led3,LOW); digitalWrite(led2,HIGH); digitalWrite(led4,HIGH); delay(500); } </pre>
--	--

Q6. What is the function of pinMode() in Arduino?

Answer:

pinMode() is used to configure a pin as **INPUT** or **OUTPUT** in Arduino.

• Reciver Code <pre>#include <SoftwareSerial.h> //XBee TX → Arduino pin 3 //XBee RX → Arduino pin 4 SoftwareSerial xbee(3, 4); void setup() { Serial.begin(9600); xbee.begin(9600);</pre>	<pre>} void loop() { if (xbee.available()) { String data = xbee.readStringUntil('\n'); data.trim(); Serial.print("Received: "); Serial.println(data); } }</pre>
--	--

Q4. Execute the program using Arduino IDE. Show the output to the examiner.

Output:

The message "**Hello**" is transmitted wirelessly and displayed on the receiver serial monitor.

Q5. List the names of short-range devices.

- Bluetooth
 - ZigBee
 - Wi-Fi
 - Infrared (IR)
 - RFID
 - NFC
-

Q6. State the applications of ZigBee.

Applications of ZigBee:

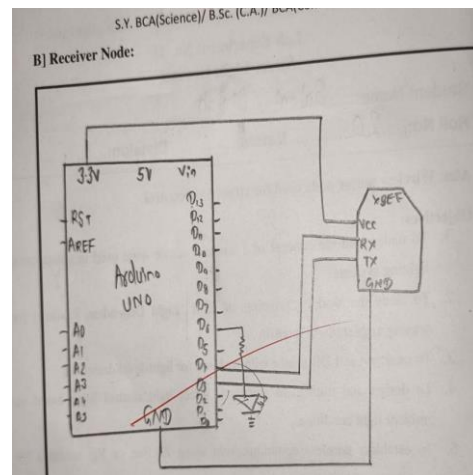
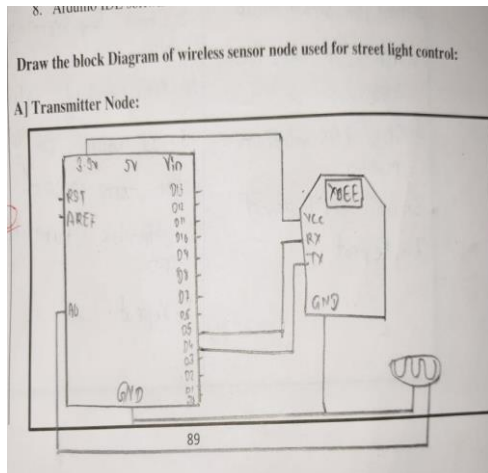
- | | |
|--|---|
| <ul style="list-style-type: none"> • Home automation • Industrial automation • Wireless sensor networks | <ul style="list-style-type: none"> • Smart energy meters • Security systems • Medical monitoring systems |
|--|---|

11. Wireless Sensor Node used for Street Light Control

Aim:

To study a wireless sensor node for automatic street light control using Arduino.

Q1. Draw the circuit / block diagram of wireless sensor node used for street light control.



Q2. Write algorithm / flowchart for same.

Algorithm:

1. Start
2. Initialize sensor and serial communication
3. Read LDR sensor value
4. Check light intensity
5. If light intensity is low, turn ON LED
6. If light intensity is high, turn OFF LED
7. Transmit sensor data through ZigBee
8. Repeat continuously
9. Stop

Q3. Write program for wireless sensor node used for street light control.

Streetlight Transmission Code

Receiver Code

Q4. Execute the program using Arduino IDE. Show the output to the examiner.

Output:

When surrounding light decreases, the LED (street light) turns ON automatically. When light increases, the LED turns OFF.

Q5. State the application of this experiment.

Applications:

- Automatic street light systems
 - Smart city projects
 - Energy-saving systems
 - Traffic monitoring systems
 - Industrial lighting control
-

Q6. State the function of Serial.begin(9600).

Answer:

Serial.begin(9600) initializes serial communication between Arduino and computer with a baud rate of **9600 bits per second**.

Observations:

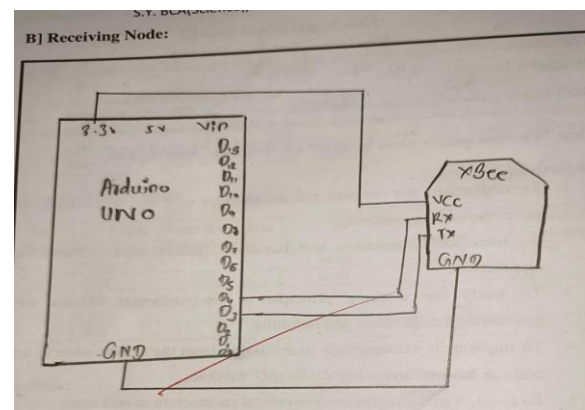
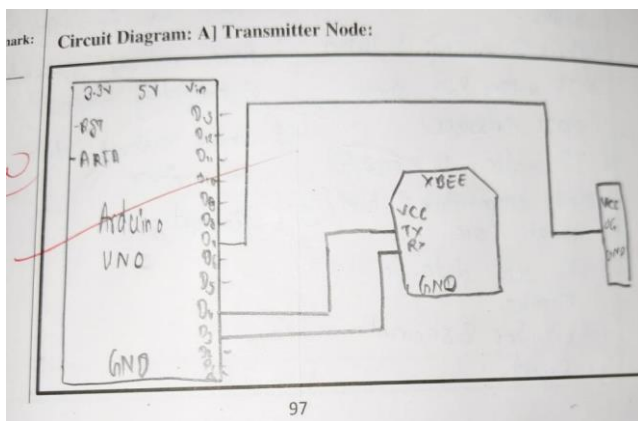
- LDR sensor detected changes in light intensity.
 - LED switched automatically according to light conditions.
 - Serial communication worked successfully.
 - Automatic street light operation was achieved.
-

12. Wireless Sensor Node used for Environment Monitoring

Aim:

To study a wireless sensor node for environment monitoring using Arduino.

Q1. Draw the circuit / block diagram of Wireless sensor node used for environment monitoring.



Q2. Write algorithm / flowchart for same.

Algorithm:

1. Start
2. Initialize sensor and serial communication
3. Read sensor values
4. Process sensor data
5. Send data through ZigBee
6. Display values on Serial Monitor
7. Repeat continuously
8. Stop

Q3. Write program for wireless sensor node used for environment monitoring.

Q4. Execute the program using Arduino IDE. Show the output to the examiner.

Output:

Sensor Value: 350

Sensor Value: 365

Sensor Value: 342

Sensor readings are displayed continuously on the Serial Monitor.

Q5. State the function of Serial.available().

Answer:

Serial.available() checks the number of bytes available to read from the serial port.

Q6. List the names of different sensors that can be used for an environment monitoring system.

- Temperature sensor
 - Humidity sensor
 - Gas sensor
 - Air quality sensor
 -
 - Light (LDR) sensor
 - Pressure sensor
 - Soil moisture sensor
 - Rain sensor
-

Observations:

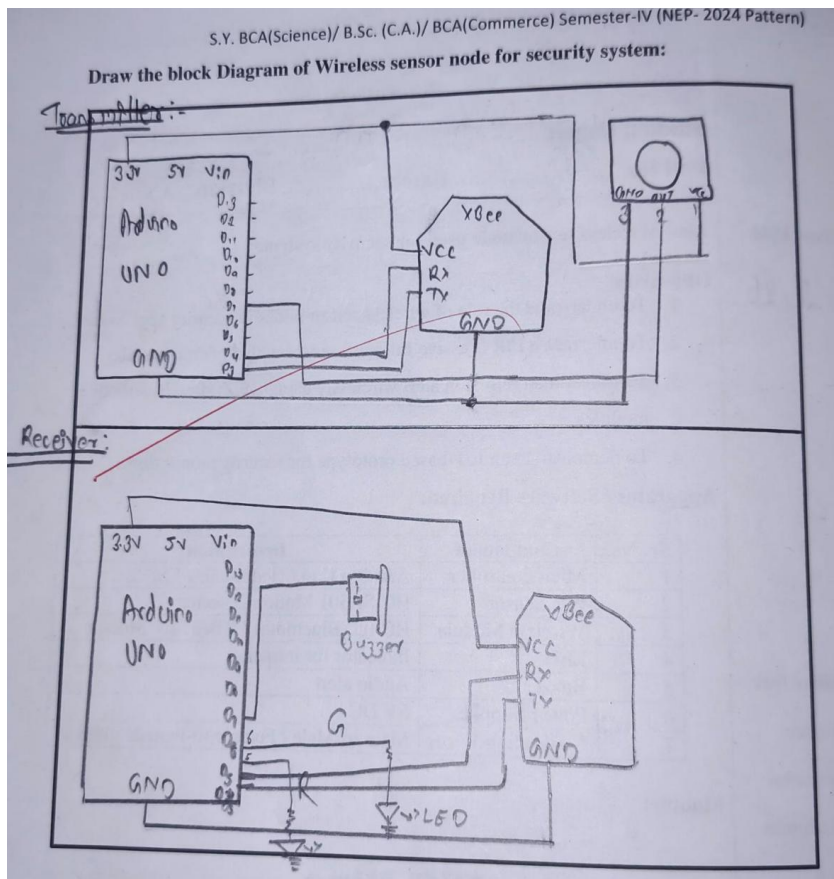
- Sensor values were read successfully.
 - Data was transmitted and displayed on the Serial Monitor.
 - Environmental parameters were monitored continuously.
-

13. Wireless Sensor Node used for Security System

Aim:

To study a wireless sensor node for a security system using Arduino and PIR sensor.

Q1. Draw the circuit / block diagram of Wireless sensor node used for security system.



Q2. Write algorithm / flowchart for same.

Algorithm:

1. Start
2. Initialize PIR sensor and serial communication
3. Read PIR sensor value
4. Check motion detection status
5. If motion detected, turn ON LED/Buzzer
6. If no motion, keep LED/Buzzer OFF
7. Send data wirelessly through ZigBee
8. Repeat continuously
9. Stop

Q3. Write program for wireless sensor node used for security system. (CODES)

Q4. Execute the program using Arduino IDE. Show the output to the examiner.

Output:

Motion Detected

No Motion

Motion Detected

LED/Buzzer turns ON when motion is detected.

Q5. State the function of digitalWrite(9,HIGH).

Answer:

digitalWrite(9,HIGH) sets pin number **9** to **HIGH state** and turns ON the connected device such as an LED or buzzer.

Q6. How does the PIR sensor detect motion?

Answer:

The PIR sensor detects motion by sensing changes in **infrared radiation (heat energy)** emitted by moving objects or humans.

Observations:

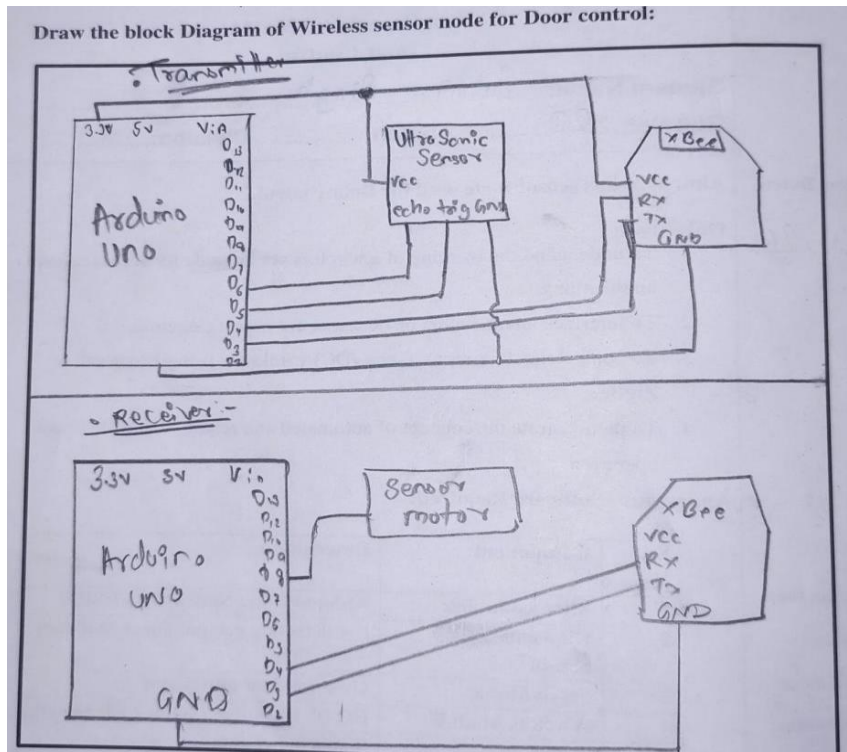
- PIR sensor detected motion successfully.
 - LED/Buzzer responded according to motion detection.
 - Sensor data was displayed on Serial Monitor.
 - Security system operated correctly.
-

14. Wireless Sensor Node used for Door Control

Aim:

To study a wireless sensor node for door control using Arduino and a servo motor.

Q1. Draw the circuit / block diagram of Wireless sensor node used for Door control.



Q2. Write algorithm / flowchart for same.

Algorithm:

1. Start
2. Initialize servo motor and serial communication
3. Read input signal from ZigBee/sensor
4. Check received command
5. If command is received, rotate servo motor to open door
6. Otherwise keep door closed
7. Repeat continuously
8. Stop

Q3. Write program for wireless sensor node used for Door control.

Q4. Execute the program using Arduino IDE. Show the output to the examiner.

Output:

Door Open

Door Closed

Door Closed

Servo motor rotates and opens/closes the door automatically

Door Open

Q5. What is the role of servo motor in this experiment?

Answer:

The servo motor is used to control the opening and closing movement of the door.

Q6. State the function of `Serial.println("Hello")`.

Answer:

`Serial.println("Hello")` displays the text **Hello** on the Serial Monitor and moves the cursor to the next line.

Observations:

- Servo motor interfaced successfully with Arduino.
 - Door opening and closing actions worked correctly.
 - Messages were displayed on the Serial Monitor.
 - Wireless door control operation was achieved.
-